

Scientific authorship in the age of collaborative research

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Abstract

I examine two challenges that collaborative research raises for science. First, collaborative research threatens the motivation of scientists. As a result, I argue, collaborative research may have adverse effects on what sorts of things scientists can effectively investigate. Second, collaborative research makes it more difficult to hold scientists accountable. I argue that the authors of multi-authored articles are aptly described as plural subjects, corporate bodies that are more than the sum of the individuals involved. Though journal editors do not currently conceive of the authors of multi-authored articles this way, this conception provides us with the conceptual resources to make sense of how collaborating scientists behave.

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1. Introduction

Collaborative research poses interesting and serious challenges for the traditional notion of the author (see Biagioli, 2003, p. 274). In some fields, concerns have been explicitly acknowledged. For example, in the biomedical sciences, some journal editors have articulated criteria for inclusion as author in papers published in their journals and demanded that prospective authors sign a statement of authorship responsibility (Biagioli,

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1999, pp. 18–20). Similarly, in high energy physics, a number of collaborative research teams and institutions have articulated policies of authorship (Galison, 2003, p. 332).¹

In this paper I want to explore the threat posed by collaborative research to scientific authorship and responsibility. In Section 2, I examine the changing role of collaborative research in science and the advantages afforded by collaboration. In Sections 3 and 4, I examine two concerns that collaborative research raises. First, in Section 3, I argue that because collaborative research threatens the motivation of scientists, it may have adverse effects on *what sorts of things* scientists can effectively investigate. Second, in Section 4, I argue that when scientists work collaboratively, it is more difficult to hold them accountable when problems arise. Consequently, I argue, collaborative research may have an adverse effect on *the quality of findings* reported.

In an effort better to understand the significant changes in scientific practice that are a consequence of collaboration, I propose that the authors of collaborative papers be conceptualized not as groups of individuals, but as plural subjects, corporate bodies that are more than the sum of the individuals involved. My proposal provides us with the conceptual resources to make sense of how collaborating scientists behave. Unfortunately, though, I argue that the reward system in science is currently not suited to deal with such authors. I argue that we need to ensure that (i) the goals of individual collaborating scientists, and (ii) the goals of collaborative research teams, coincide with (iii) the goals of science. Only then can we expect to exploit the many benefits afforded by collaborative research, without incurring significant costs.²

2. Collaborative research: a brief history

The term ‘collaborative research’ is used in a variety of ways (see Thagard, 1999). Some, for example, have suggested that *all* people referred to in the acknowledgement of an article are collaborators (see Cronin, Shaw, & La Barre, 2003). I will use the term ‘collaborative research’ narrowly to designate research that results in co-authored publications. And, throughout this paper, I will count as collaborators only those whose names are listed as authors.

¹ As Peter Galison (2003) notes, one of the concerns in the debates in high-energy physics is to determine who can count as an author. Technicians, for example, play an indispensable role in the construction and maintenance of the equipment and instruments used in high-energy physics. But, the traditional conception of scientific authorship seems to preclude their being counted as authors.

² Collaboration is not the only new practice in science that challenges traditional conceptions of authorship. Ghost-writing has raised serious challenges in medical research, a field in which funding is often provided by private sector pharmaceutical companies (see Lynch, 2004, Sismondo, 2004, and Healy, 2004). Such companies are willing and able to finance the publication of research. And they have strong incentives to look for people with impressive academic credentials who will stand behind their findings. As Michael Lynch (2004, p. 147) notes, pharmaceutical companies have even presented ‘prospective “authors” with draft reports of “their own” research’. For many scientists, the temptation to accept authorship for a paper one has not written can be too great to resist. After all, research careers are measured to a large degree in terms of publications. And some researchers may feel obliged to accept such offers from sources that have funded their research in the past. Sergio Sismondo (2004, p. 151) notes that as ghost-writing becomes more prevalent, we have reason to question whether or not authorship is at all connected ‘with the research or writing that’ produces scientific publications. The practice of ghost-writing is not new to science. As Steven Shapin (1994, pp. 356–358) notes, Denis Pepin was a ghost-writer for Robert Boyle. That is, Pepin wrote articles for which Boyle was given authorial credit. There are significant differences between Pepin and these contemporary cases, though.

Such research was relatively uncommon for much of the history of modern science. Collaborative research accounted for only about 25% of the published papers in the natural sciences between 1900 and 1909 (Zuckerman & Merton, 1973, p. 547). Throughout the twentieth century things changed dramatically. First in physics and astronomy, and then in the biomedical sciences, collaborative research became increasingly popular. By the 1960s, collaborative research accounted for over 80% of the papers published in the natural sciences (*ibid.*). Consulting the *Web of Science* I found that in some journals up to 88% of the articles are co-authored (Wray, 2002, pp. 151–152). The *Journal of the American Chemical Society* is perhaps one of the most extreme cases. Since the 1970s, over 94% of the articles in this journal have been co-authored, and throughout the 1990s 98% were co-authored (Cronin, Shaw, & La Barre, 2004, p. 165). Collaboration is also now the norm in some of the leading journals in the social sciences. For example, in *Psychological Review*, ‘during the nineties, 71% of all research articles were coauthored’ (Cronin et al., 2003, p. 861).

Increasingly, papers are being authored by greater numbers of authors. By the 1990s the mean number of authors for articles published in *Physical Review Letters* was 5.5 (Thagard, 1999, p. 169). ‘In the *Journal of Neurosurgery* and *Neurosurgery* ... the average number of authors per scientific article rose from 1.8 to 4.6’ between 1945 and 1995 (Cronin, 2002). There are papers in the physical sciences that list over fifty authors (Hardwig, 1985; Beaver & Rosen, 1979). Indeed, the number of papers with 100 or more authors has increased ‘from 1 in 1981 to 182 in 1994’ (Cronin, 2002). Even in philosophy of science we now see more papers authored by more than three authors than we did thirty years ago (Wray, 2002).

In many respects, the move toward collaborative research has been good for science. Elsewhere, I have identified a number of benefits of collaborative research (*ibid.*). It is worth briefly reviewing these. First, as numerous studies confirm, generally collaborative papers are of a higher quality as indicated by how frequently they are cited (*ibid.*, p. 156; Frank Fox, 1991). This conclusion has been further confirmed in a recent study by Don Beaver (2004).³

Second, there are some things that could only be discovered collaboratively. Some research depends upon scientists bringing together a variety of knowledge from different fields, knowledge that no single scientist is apt to have. Paul Thagard (1999) cites the example of the development of the bacterial theory of ulcers. Different scientists often contribute different skills to a research project. For example, Mario Biagioli (1999, p. 17) cites cases of research projects in biomedicine that required the assistance of biostatisticians. Moreover, there are investigations in particle physics that require the concerted efforts of many scientists over many years (Hardwig, 1985; Beaver, 2001, p. 369). Such investigations would be unthinkable if scientists worked independently.

Third, collaborative research ensures that findings are less apt to be forgotten (Wray, 2002; Merton, 1973). The vast quantity of scientific information available today has raised serious challenges to those seeking to find information (see Price, 1986, p. 7). As a result, many past discoveries that have important implications for current research may be neglected. Perhaps the most famous case of this is Mendel’s discovery. Initially his discov-

³ Beaver (2001, p. 367) also notes that ‘core journals have higher frequencies of collaborative papers than the average journal’. As Beaver (*ibid.*, p. 367) explains, ‘core journals form a small yet visible elite, and as such, display characteristics of the scientific elite’.

ery was largely neglected. Consequently, the key insight associated with Mendel had to be *rediscovered* years later (see Brannigan, 1981).⁴ But, when research is done collaboratively, there are more minds consciously aware of *what* has been discovered and *where* desired information can be found.⁵

Fourth, collaborative research plays an important role in the training of young scientists (Wray, 2002; Thagard, 1999). Young scientists often learn the norms of science from working closely and collaboratively with mentors who are established in their field. Harriet Zuckerman (1996), for example, notes that Nobel laureates, the elite of the elite in the natural sciences, are especially apt to have collaborated with a mentor early in their training, often with a scientist who either has been or later was honored with a Nobel prize (see also Beaver, 2001, p. 372).

Thus, collaboration is an important feature of contemporary science both because it is so pervasive, and because of the benefits it has brought to science.

3. The costs of collaboration: reduced motivation

I now want to examine two concerns that collaborative research raises. The first concern is that collaborative research seems to threaten the motivation of scientists. This, in turn, may have adverse effects on what sorts of things scientists can effectively investigate.

There is some evidence that scientists who work collaboratively are worrying increasingly about getting adequate credit for their contributions to collaborative projects. Linda Wilcox (1998, p. 216), for example, notes that at the Harvard Medical School, Dental School, School of Public Health, and affiliated hospitals ‘queries related to authorship have grown as a percentage of the total complaints to the Ombuds Office’. Wilcox (*ibid.*) found that ‘from the 1991–1992 through the 1996–1997 academic years’ such queries grew ‘from 8 (2.3%) to 59 (10.7%)’. And, in a survey of biomedical practitioners conducted at the University of Newcastle, 49% of the respondents claimed to have ‘experienced situations in which ... authorship had been deserved but not awarded’ (Biagioli, 1999, p. 22). Similarly, ‘when the European Committee for Future Accelerators polled members of the big teams in 1995, they found that some seventy-five percent of the respondents disliked present publication habits, largely because it damaged the possibility of career advancement and of receiving credit for their work’ (Galison, 2003, p. 341). Thus, scientists are increasingly expressing concerns that they are not being given adequate credit for work they do on collaborative projects. In some fields this concern is now explicitly acknowledged. As Peter Galison (2003, p. 336) notes, at some research institutions specializing in high-energy physics authorship policies have been designed with an aim to ensuring that all contributors to a published research project are given recognition.

If scientists worry about getting their due credit when they work collaboratively, there is apt to be tension between those working together. As John Heilbron (1992, p. 46) explains, in collaborative projects in high-energy physics ‘individuals must play for the team while simultaneously distinguishing themselves for leadership and good taste’. Heil-

⁴ Brannigan (1981) suggests that the significance that Mendel’s findings have for understanding inheritance even escaped Mendel’s attention.

⁵ A related advantage of collaborative research is that ‘errors are more readily detected when several different individuals with different perspectives discuss ... data and ... theory’ (Beaver, 2001, p. 370).

bron believes that accelerator and detector *builders* face a real challenge because they have ‘fewer chances to call attention to themselves than have experimenters’ (ibid.). In situations such as these, where a scientist feels that she may not get the credit she feels she deserves, tensions are inevitable.

As both Robert Merton (1973) and David Hull (1988) note, the success of science depends to some extent on scientists getting credit for the work they produce. The recognition that a scientist gets from her peers for making a significant discovery is a key motivation for scientists. Indeed, to some degree, the reliability of scientific findings is a function of the credit system in science. Knowing that one stands either (i) to gain credit for making an important contribution, or (ii) to lose credibility if one’s findings later prove to be unreliable, scientists are motivated to produce results that are generally reliable (see Hull, 1988).

As Zuckerman & Merton (1973) note, the anxiety that scientists have about getting credit for their contributions to collaborative projects is apt to erode their motivation. If a scientist cannot be confident that she will get credit when she contributes to a collaborative project, her motivation to work on such projects is reduced markedly. This could have a negative impact on the sorts of research projects in which scientists choose to engage. If scientists have anxiety about working on collaborative projects, and some parts of nature can only be investigated effectively if scientists work collaboratively, we may find that some parts of nature elude our understanding. Indeed, if the anxiety scientists feel about getting credit on collaborative projects increases, we may find that scientists are reluctant to pursue training in certain fields for fear that they will end up working on projects for which their chance of receiving credit is low.

In summary, collaborative research has created a tension in science that threatens to restrict our understanding of the world. On the one hand, unless scientists work collaboratively, there are parts of the world that we cannot know. On the other hand, sometimes the reward system in science does not give scientists adequate incentives to work collaboratively. The risks that scientists take when they choose to work collaboratively do not affect all parties equally. Consequently, because of the way the reward system works, scientists lack incentives to investigate certain parts of the world, parts that can only be effectively investigated collaboratively.

4. The costs of collaboration: reduced accountability

I now want to consider the second concern, that collaborative research may have an adverse effect on the reliability of the research produced. As Steven Shapin (1994) notes, trust is inevitable in a social system as complex as science (see also Biagioli, 2003, p. 260). Scientists need to trust that the data their peers report are in fact reliable. Though scientists do *sometimes* take the time to check the findings reported by others, more often they trust what is reported by their peers. And, scientists routinely employ instruments that were developed by others who depended upon theories developed in other fields, fields that the one scientist relying on the instrument may understand only vaguely (Duhem, 1996 [1894], pp. 81–87; Latour & Woolgar, 1986, pp. 63–64). It is in this respect that trust is the foundation of scientific knowledge (Shapin, 1994). And, the credit system in science needs to be structured to ensure that scientists can generally trust the findings reported by their peers.

But, collaborative research can result in publications for which the expected trust is unfounded. A scientist can participate in a collaborative research project and be quite unaware of the contributions of other participants. Biagioli (1999, p. 19) notes cases of senior scientists who were listed as authors on papers that were subsequently found to contain mistakes who pleaded innocence on the grounds that ‘they were not responsible for the mistakes and misconduct of their junior colleagues’. A scientist can also find her own contributions to a project used or interpreted in ways that she does not *personally* accept. Further, she may find herself listed as an author for an article that she understands only partially (see *ibid*, p. 17). As Biagioli (*ibid*, p. 22) notes, scientists do not believe ‘they can or should vouch for’ the contributions of those with whom they collaborate.

In order to make sense of the behavior of scientists when they work collaboratively, we need to rethink our understanding of scientific authorship. The co-authored article is not aptly described as authored by two or more *individual* authors. Rather, scientists working together on collaborative projects are aptly described as forming a plural subject. A plural subject is an agent composed of individuals that has an existence above and beyond the constitutive individuals (see Gilbert, 1992). The plural subject has intentions that *may* differ from the intentions of the constitutive individuals. Many of us have, at one time or another, participated in plural subjects. For example, one may have been part of a committee that made a decision that one was against personally. Still, the decision stood as the decision of the *committee*. The decisions of plural subjects *need not* conflict with the preferences of the constitutive members. The point is that they can and sometimes do.

A collaborative research team creates an author that is, in some sense, above and beyond the sum of its parts.⁶ For example, when Watson and Crick set out to author an article together, a new author emerged, one not completely reducible to the two individual authors, James Watson and Francis Crick. Their collaboration gave rise to thoughts and abilities that were not available to each working independently. Others have noted the complex nature of collaborative research teams. For example, Ron Giere (2002, pp. 640–641) argues that in a collaborative research team cognition is distributed. That is, rather than being localized in the individual scientists involved, the cognition of such groups is aptly described as extending across the individuals, even embracing the instruments and equipment they employ. The plural subject conception of authorship I am proposing here draws on these insights.

Given that *the research team* is the author of the findings reported, it is the team that is the entity who should be held responsible. But, once we conceptualize the authors of collaborative research as plural subjects we can understand why these ‘authors’ are not as responsible as traditional authors. Given the nature of plural subjects, group responsibility can be quite elusive. Margaret Gilbert’s (2000) account of group responsibility offers insight into why there is a misfit between collaborative research and authorial responsibility. As Gilbert explains, in order for a group to be held responsible for an error the following three conditions need to be met: (1) a group must act, (2) it must act freely, and

⁶ Beaver (2001, p. 371) makes a similar suggestion. As he explains, ‘each research group could be seen as a kind of “person”, or “individual”, just as in (American) law, a corporation as a legal entity is a person or being’.

(3) it must accept that a certain action is wrong. Plural subjects, including collaborative research teams, often act as groups, and do so freely.⁷ Hence, conditions (1) and (2) are often met.

But condition (3) is more problematic. Though *individual members* of a collective may readily admit problems with some action of the group, it is quite a different matter to get the *plural subject* to make such an admission. Consequently, when there are problems with the results reported in an article that is collaboratively authored, or even just suspicion that there may be a problem, it can be quite difficult to get the research team to acknowledge this. Fearing that *individually* they may be held responsible, the various individual members of a collaborative research team may have incentives to prevent the group from acknowledging a problem. Provided *the group* does not admit wrongdoing it will be difficult to hold the individual members of the group accountable. Sometimes scientists who have worked collaboratively can honestly deny that they are culpable when an article they worked on is found to contain errors, for sometimes they are unaware of the use to which their contribution is put.

In summary, it seems that collaborative research gives rise to a paradox. Collaborative research depends upon trust. But, such research often seems to erode the grounds of trust that has made science work so well in the past. Hence, though collaboration opens up new areas for research that are inaccessible to those who do not collaborate, it also threatens to undermine our confidence in the findings produced.

In scientific fields in which collaborative research is the norm we need some means to ensure that scientists both remain motivated, and are held responsible. As Hull (1988) notes, the success of science has been achieved, at least in part, by a reward system that ensured that the goals of individual scientists and the goals of science, the institution, coincide. The goals of science, the institution, include the development of models and theories that accurately reflect features and processes in the world. And, though individual scientists also often have this goal in mind, they also have other goals, such as career advancement, proving themselves to their supervisors, or getting another publication. Hence, the goals of individual scientists may conflict with the goals of science. In the past, individual scientists have generally been able to advance the goals of science by acting in their own interests.

The situation we face in the age of collaborative research is more complex. We now have more potentially competing goals to coordinate. In addition to (i) the institutional goals of science, and (ii) the goals of individual scientists, we also have to consider (iii) the goals of research teams. For example, in the early 1950s, one of the goals of science, the institution, was to develop an understanding of the structure of DNA. One of goals of the research team composed of Watson and Crick was to avoid the mistakes of Linus Pauling's laboratory team. And, one of the goals of Watson may have been to land a permanent job at Harvard.

⁷ A group's properties are not necessarily the same as the properties of the individual members. Hence, a group, qua group, can act freely even when the individual members of the group are coerced to act as they do. Consequently, even though a research team working in a laboratory together may operate in an undemocratic fashion, with the principal investigator calling all the shots, the *group* can still act freely. Whether or not the group acts freely is a function of whether or not *the group* is coerced, not whether or not the individual members of the group are coerced. I thank one of the referees for *Studies in History and Philosophy of Science* for pushing me to clarify this point.

In the age of collaborative research, there are new types of conflicts that may arise. The goals of the research team may conflict with the goals of the individual members of the team. That is, in Watson's pursuit of a job at Harvard, Watson and Crick could have been led to make the same mistakes Pauling made. Similarly, the goals of a research team can also conflict with the institutional goals of science. That is, sometimes, in their effort to advance their collective goals, Watson and Crick may have acted in ways that were an impediment to the advancement of the institutional goals of science. For example, they may have withheld publishing or sharing information that would have been useful to other scientists working on the problem, including Pauling and Rosalind Franklin.

The challenge we now face is to restructure the reward system in science in such a way as to ensure the convergence of (i) the goals of individual researchers, (ii) the goals of research teams, and (iii) the goals of science. Importantly, it is not necessary that the goals be the same. Rather, it is sufficient that the various goals lead to the same result.

5. Concluding remarks

In summary, collaborative research benefits science in a variety of ways. It enables scientists to investigate parts of the world otherwise beyond our understanding. It provides a means to ensure that past findings are less apt to be forgotten. It plays an important role in the training of scientists. And, in *some* respects, it seems to increase the quality of research produced.

But, I have argued that collaborative research also raises concerns. First, collaborative research seems to threaten the motivation of scientists, which may have adverse effects on the sorts of things scientists can effectively investigate. Second, when scientists work collaboratively, it is more difficult to hold them accountable when problems arise, which may have an adverse effect on the quality of findings reported. I developed a plural subject account of scientific authorship to account for the behavior of scientists working collaboratively.

Currently, journal editors do not regard the authors of collaborative research as plural subjects. Rather, as Biagioli (1999, p. 23) notes, the '[International Committee of Medical Journal Editors]' guidelines break multiauthorship down to an assembly of separate authors, each fully individual and fully responsible', with the consequence that 'coauthorship cannot mean corporate authorship'. But, as I have argued, the traditional individualist conception of authorship is descriptively inadequate for understanding what currently happens in collaborative research projects. Consequently, if we want to continue to rely on the reward system to work as the guarantor of the findings reported in collaborative research projects, the reward system in science must be altered to acknowledge that this work is done by plural subjects.

The task of redesigning the reward system in science extends beyond the scope of this paper. The current reward system in science developed and grew with our conception of the lone author. The current reward system in science and the lone author fit together quite nicely. But, the plural subject author who now dominates in some fields of science does not respond and act as a traditional lone author. In redesigning the reward structure a variety of goals must be made to converge: (i) the goals of individual researchers who work collaboratively, (ii) the goals of collaborative research teams, and (iii) the institutional goals of science.

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